

```

''' r
#Load data
turnover <- read_csv("turnover.csv")
glimpse(turnover)
'''

'''
## Rows: 46,710
## Columns: 6
## $ date      <date> 1982-04-01, 1982-04-01, 1982-04-01, 1982-04-01, 1982-0~
## $ value     <dbl> 303.1, 41.7, 408.7, 211.3, 94.0, 178.3, 50.4, 178.0, 85~
## $ state     <chr> "New South Wales", "New South Wales", "New South Wales"~
## $ industry_group <chr> "Supermarket and grocery stores", "Liquor retailing", "~
## $ year      <dbl> 1982, 1982, 1982, 1982, 1982, 1982, 1982, 1982, 1982, 1~
## $ month     <dbl> 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4~
'''

''' r
#Check for missing data
turnover %>%
summarise(
  missing_date = sum(is.na(date)),
  missing_value = sum(is.na(value)),
  missing_state = sum(is.na(state)),
  missing_industry = sum(is.na(industry_group)),
  missing_year = sum(is.na(year)),
  missing_month = sum(is.na(month))
)
'''

'''
## # A tibble: 1 x 6
##   missing_date missing_value missing_state missing_industry missing_year
##       <int>         <int>         <int>           <int>         <int>
## 1             0         3878             0             0             0
## # i 1 more variable: missing_month <int>
'''

```

- The dataset has 46,710 rows and six variables. **date** is stored as a date variable; **value**, **year** and **month** are numerical variables; **state** and **industry_group** are character variables. Only **value** has 3,878 missing values. The missing turnover values were kept instead of replaced or deleted because changing them could lead to bias or false information. Below is the description of each variable:

```

# 6 variables

#1. Variable 1: Date
#a. Date summary

#group data by date
by_date <- turnover %>%
group_by(date) %>%
summarise(number_of_observations = n())

head(by_date)

```

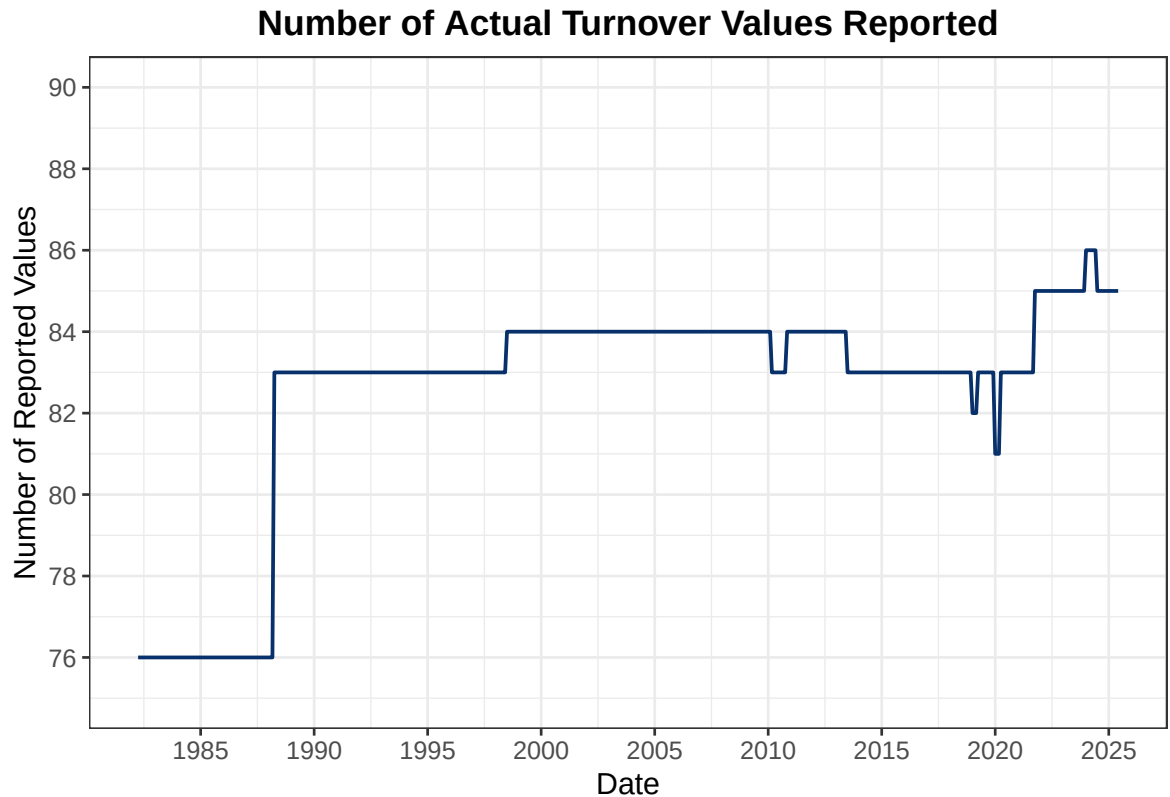
```
## # A tibble: 6 x 2
##   date      number_of_observations
##   <date>          <int>
## 1 1982-04-01             90
## 2 1982-05-01             90
## 3 1982-06-01             90
## 4 1982-07-01             90
## 5 1982-08-01             90
## 6 1982-09-01             90
```

```
#summarise date range
by_date %>%
summarise(
  first_date = min(date),
  last_date = max(date),
  number_of_months = n(),
  minimum_observations_per_month = min(number_of_observations),
  maximum_observations_per_month = max(number_of_observations)
)
```

```
## # A tibble: 1 x 5
##   first_date last_date  number_of_months minimum_observations_per_month
##   <date>    <date>          <int>                <int>
## 1 1982-04-01 2025-06-01             519                  90
## # i 1 more variable: maximum_observations_per_month <int>
```

```
#b. Data availability over time
reported_by_date <- turnover %>%
group_by(date) %>%
summarise(reported_values = sum(!is.na(value)),
  .groups = "drop")

#plotting
ggplot(reported_by_date,aes(x = date, y = reported_values)) +
  geom_line( colour = "#08306B",linewidth = 0.7) +
  labs(
title = "Number of Actual Turnover Values Reported",
x = "Date",
y = "Number of Reported Values") +
  scale_x_date( date_labels = "%Y", date_breaks = "5 years" ) +
  scale_y_continuous(breaks = seq(76, 90, by = 2)) +
  coord_cartesian(ylim = c(75, 90)) +
  theme_bw() +
  theme(
text = element_text(size = 12),
plot.title = element_text(face = "bold", hjust = 0.5))
```



- **date:** Illustrates when turnover data were recorded. All dates use the first day of the month as a standard monthly reference date. For example, 1982-04-01 represents the month of April 1982. The dataset has a total of 519 months, from April 1982 to June 2025. Each date has 90 rows, representing nine states (including total states) combined with ten industry series (including total industries). Data availability generally improves over time, although temporary decreases occur around 2010, 2013, 2019 and 2020. “Number of Reported Turnover Values Each Month” graph measures data completeness and should not be interpreted as showing changes in the amount of turnover.

```
# Variable 2: State
```

```
#count the observations and arrange states alphabetically
by_state <- turnover %>%
group_by(state) %>%
summarise(number_of_observations = n(),
.groups = "drop") %>%
arrange(state)

by_state
```

```
## # A tibble: 9 x 2
##   state                                number_of_observations
##   <chr>                                <int>
## 1 Australian Capital Territory          5190
## 2 New South Wales                      5190
## 3 Northern Territory                  5190
## 4 Queensland                          5190
## 5 South Australia                      5190
```

```
## 6 Tasmania 5190
## 7 Total (State) 5190
## 8 Victoria 5190
## 9 Western Australia 5190
```

- **state:** Illustrates where each turnover observation belongs to. It includes eight Australian states and territories plus Total (State), giving nine categories. Although there are 5,190 rows for each state, it doesn't mean that they have equal turnover or equal number of reported turnover values because `n()` also counts rows where value is missing.

#Variable 3: Industry

```
#count the observations and arrange industry alphabetically
by_industry <- turnover %>%
group_by(industry_group) %>%
summarise(number_of_observations = n(),
.groups = "drop") %>%
arrange(industry_group)

by_industry
```

```
## # A tibble: 10 x 2
##   industry_group number_of_observations
##   <chr>          <int>
## 1 Clothing retailing 4671
## 2 Department stores 4671
## 3 Food retailing 4671
## 4 Household goods retailing 4671
## 5 Liquor retailing 4671
## 6 Newspaper and book retailing 4671
## 7 Other retailing 4671
## 8 Supermarket and grocery stores 4671
## 9 Takeaway food services 4671
## 10 Total (Industry) 4671
```

- **industry_group:** Illustrates the industry associated with each observation, including nine industries plus Total (Industry). Each category has 4,671 rows.

#Variable 4: Year

```
#number of rows each year
by_year <- turnover %>%
group_by(year) %>%
summarise(number_of_observations = n(),
.groups = "drop")

#summarise year range
year_summary <- by_year %>%
summarise(
  first_year = min(year),
  last_year = max(year),
  number_of_years = n(),
```

```

minimum_observations = min(number_of_observations),
maximum_observations = max(number_of_observations))

year_summary

## # A tibble: 1 x 5
##   first_year last_year number_of_years minimum_observations maximum_observations
##   <dbl>      <dbl>         <int>          <int>          <int>
## 1      1982      2025           44             540            1080

# Rank years from the smallest to largest number of observations
by_year_ranked <- by_year %>%
  arrange(number_of_observations, year)

print(by_year_ranked, n = 44)

## # A tibble: 44 x 2
##   year number_of_observations
##   <dbl>          <int>
## 1 2025             540
## 2 1982             810
## 3 1983            1080
## 4 1984            1080
## 5 1985            1080
## 6 1986            1080
## 7 1987            1080
## 8 1988            1080
## 9 1989            1080
## 10 1990            1080
## 11 1991            1080
## 12 1992            1080
## 13 1993            1080
## 14 1994            1080
## 15 1995            1080
## 16 1996            1080
## 17 1997            1080
## 18 1998            1080
## 19 1999            1080
## 20 2000            1080
## 21 2001            1080
## 22 2002            1080
## 23 2003            1080
## 24 2004            1080
## 25 2005            1080
## 26 2006            1080
## 27 2007            1080
## 28 2008            1080
## 29 2009            1080
## 30 2010            1080
## 31 2011            1080
## 32 2012            1080
## 33 2013            1080
## 34 2014            1080

```

```
## 35 2015 1080
## 36 2016 1080
## 37 2017 1080
## 38 2018 1080
## 39 2019 1080
## 40 2020 1080
## 41 2021 1080
## 42 2022 1080
## 43 2023 1080
## 44 2024 1080
```

```
#check the two partial years at the beginning and end of the dataset
turnover %>%
  filter(year %in% c(1982, 2025)) %>%
  group_by(year) %>%
  summarise(
    first_date = min(date),
    last_date = max(date),
    number_of_months = n_distinct(month),
    number_of_observations = n())
```

```
## # A tibble: 2 x 5
##   year first_date last_date number_of_months number_of_observations
##   <dbl> <date>      <date>          <int>                <int>
## 1  1982 1982-04-01 1982-12-01             9                  810
## 2  2025 2025-01-01 2025-06-01             6                  540
```

- **year:** Illustrates only the year of each observation. It has 44 years, from 1982 to 2025. Most years were fully recorded with 12 months, returning 1,080 rows but the year of 1982 has 810 rows because the data begin in April, while 2025 has 540 rows because the data end in June. Therefore, the lower counts in these years reflect incomplete records, not lower turnover.

#Variable 5: Month

```
#count and arrange rows each month
by_month <- turnover %>%
  group_by(month) %>%
  summarise(number_of_observations = n(),
    .groups = "drop") %>%
  arrange(month)

print(by_month)
```

```
## # A tibble: 12 x 2
##   month number_of_observations
##   <dbl>                <int>
## 1     1                 3870
## 2     2                 3870
## 3     3                 3870
## 4     4                 3960
## 5     5                 3960
## 6     6                 3960
## 7     7                 3870
```

```
## 8      8      3870
## 9      9      3870
## 10     10     3870
## 11     11     3870
## 12     12     3870
```

- **month:** Illustrates 12 months in a year. Notably, this variable groups the same month together, regardless of years. April, May and June each have 3,960 rows, while every other month has 3,870 rows. This difference does not demonstrate seasonal patterns because the dataset begins in April 1982 and ends in June 2025.

#Variable 6: Value

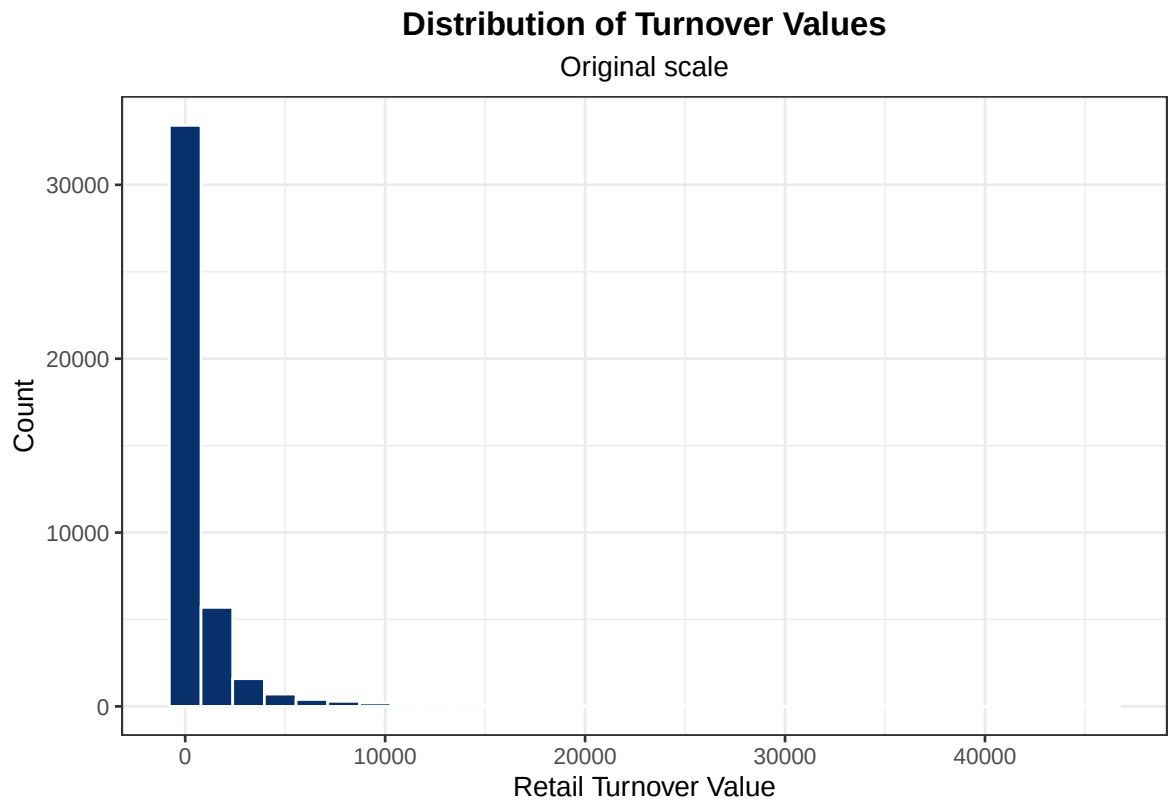
#Summarise the reported turnover values while keeping missing rows in the data

```
value_summary <- turnover %>%
summarise(
  total_observations = n(),
  reported_values = sum(!is.na(value)),
  missing_values = sum(is.na(value)),
  minimum = min(value, na.rm = TRUE),
  median = median(value, na.rm = TRUE),
  average = mean(value, na.rm = TRUE),
  maximum = max(value, na.rm = TRUE),
  standard_deviation = sd(value, na.rm = TRUE))

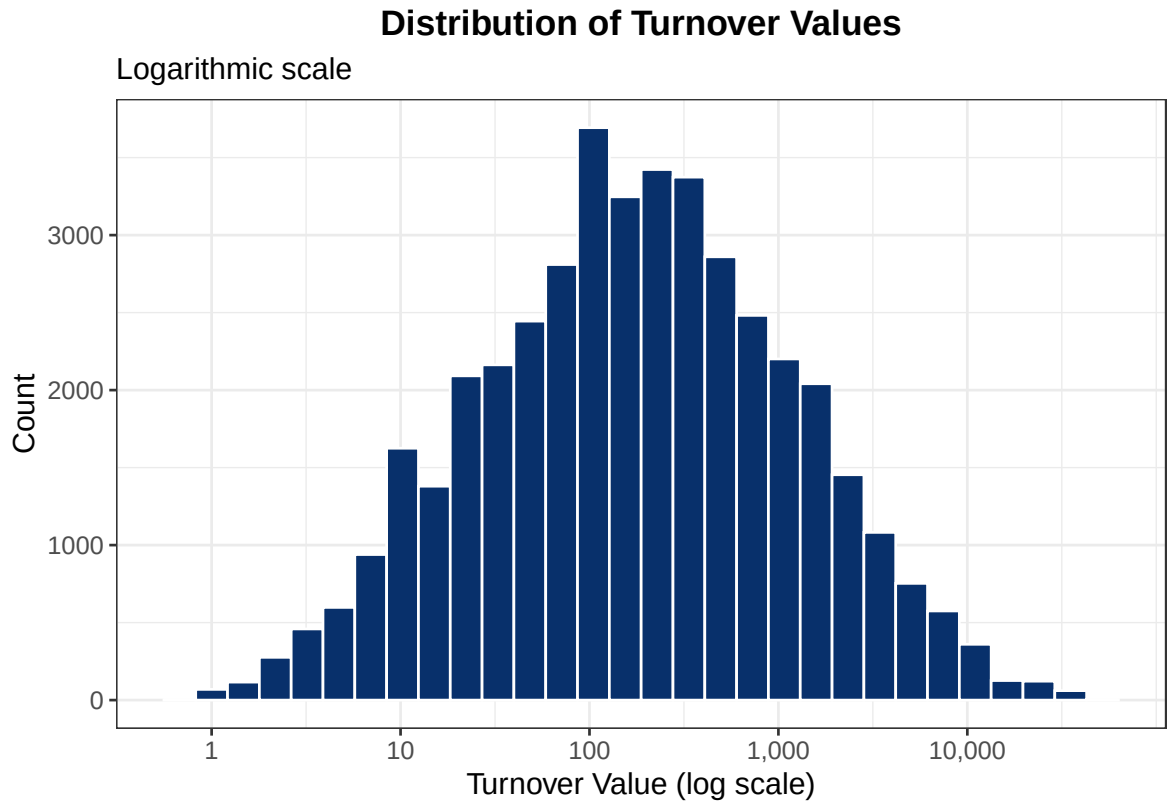
value_summary
```

```
## # A tibble: 1 x 8
##   total_observations reported_values missing_values minimum median average
##           <int>           <int>         <int>    <dbl>  <dbl>   <dbl>
## 1           46710           42832           3878     0.6   177.   894.
## # i 2 more variables: maximum <dbl>, standard_deviation <dbl>
```

```
#plotting - dollar scale
ggplot(turnover, aes(x = value)) +
  geom_histogram(
    bins = 30,
    fill = "#08306B",
    colour = "white",
    na.rm = TRUE) +
  labs(
    title = "Distribution of Turnover Values",
    subtitle = "Original scale",
    x = "Retail Turnover Value",
    y = "Count") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", hjust=0.5),
        plot.subtitle = element_text(hjust = 0.5))
```



```
#plotting - log scale
ggplot(turnover, aes(x = value)) +
  geom_histogram(
    bins = 30,
    fill = "#08306B",
    colour = "white",
    na.rm = TRUE) +
  scale_x_log10(
    breaks = c(1, 10, 100, 1000, 10000),
    labels = scales::label_comma()) +
  labs(
    title = "Distribution of Turnover Values",
    subtitle = "Logarithmic scale",
    x = "Turnover Value (log scale)",
    y = "Count") +
  theme_bw() +
  theme(
    text = element_text(size = 12),
    plot.title = element_text(face = "bold", hjust = 0.5))
```



- value:** Illustrates monthly turnover, measured in millions dollars according to the ABS description. It has 46,710 observations, in which 42,832 have reported turnover values and 3,878 are missing. The reported values range from \$0.6 million to \$45,943 million, with a median of \$176.7 million and an average of around \$894 million. The average is much higher than the median, highlighting that the distribution is strongly right-skewed. The reason behind this may be that there is a small number of observations, such as totals and turnover from larger economic states or industries, have much higher values. Growth in spending over time may also result in these high values. Therefore, a logarithmic scale is also used to show the lower and middle values more clearly without removing or changing any reported values.

```
#filter data
state_turnover_2024 <- turnover %>%
filter(
  year == 2024,
  industry_group == "Total (Industry)",
  state != "Total (State)") %>%

#Calculate annual turnover for each and its percentage
group_by(state) %>%
summarise( annual_turnover = sum(value),
  .groups = "drop") %>%
mutate(percentage = 100 * annual_turnover / sum(annual_turnover))

#rank from highest
state_turnover_2024 <- state_turnover_2024 %>%
arrange(desc(annual_turnover))

state_turnover_2024
```

```
## # A tibble: 8 x 3
##   state          annual_turnover percentage
##   <chr>          <dbl>         <dbl>
## 1 New South Wales    135639.      31.1
## 2 Victoria          112611      25.8
## 3 Queensland         89987.      20.6
## 4 Western Australia  49491.      11.3
## 5 South Australia   28136.       6.44
## 6 Tasmania          8717.       2.00
## 7 Australian Capital Territory 8150.       1.87
## 8 Northern Territory  4026.       0.922
```

```
#prepare slice positions and labels
donut_summary <- state_turnover_2024 %>%
  arrange(desc(state)) %>%
  mutate(
    label_position = cumsum(percentage) - 0.5 * percentage,
    percentage_label = paste0(round(percentage, 1), "%")
  )

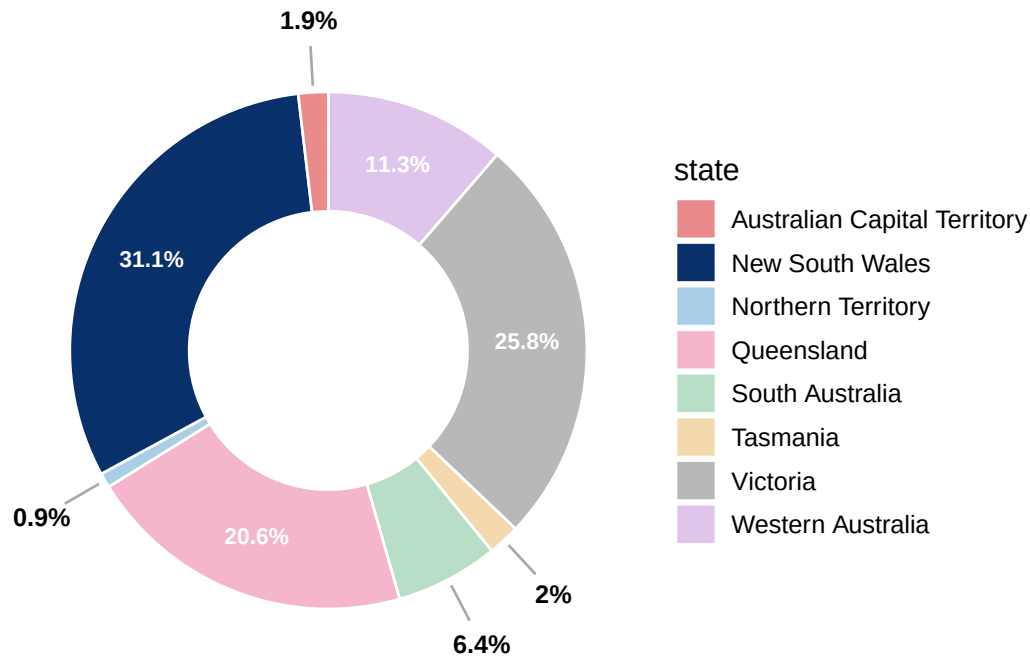
#plotting
ggplot(donut_summary,
  aes(x = 2, y = percentage, fill = state)
) +
  geom_bar(stat = "identity", colour = "white"
) +
  geom_text(
    data = donut_summary %>%
      filter(percentage >= 7),
    aes(x = 2,
      y = label_position,
      label = percentage_label),
    inherit.aes = FALSE,
    colour = "white",
    fontface = "bold",
    size = 3
  ) +
  geom_segment(
    data = donut_summary %>%
      filter(percentage < 7),
    aes(
      x = 2.5,
      xend = 2.8,
      y = label_position,
      yend = label_position),
    colour = "darkgrey",
    linewidth = 0.5
  ) +
  geom_text(
    data = donut_summary %>%
      filter(percentage < 7),
    aes(
```

```

x = 3,
y = label_position,
label = percentage_label),
  inherit.aes = FALSE,
  colour = "black",
  fontface = "bold",
  size = 3.5
) +
coord_polar(
  theta = "y",
  start = 0,
  clip = "off"
) +
scale_fill_manual(
  values = c(
    "New South Wales" = "#08306B",
    "Victoria" = "#B8B8B8",
    "Queensland" = "#F5B6CD",
    "Western Australia" = "#DFC5EB",
    "South Australia" = "#B9DEC7",
    "Tasmania" = "#F5D9AE",
    "Australian Capital Territory" = "#E98B8B",
    "Northern Territory" = "#A9CFE8")
) +
scale_x_continuous(limits = c(0.5, 2.5), oob = scales::oob_keep)
) +
labs(
  title = "Annual Turnover by State and Territory, 2024",
) +
theme_void() +
theme(
  text = element_text(size = 12),
  plot.title = element_text(face = "bold", margin = margin(b = 15)),
  legend.title = element_text(size = 12),
  legend.text = element_text(size = 10),
  plot.margin = margin(10, 30, 10, 30)
)

```

Annual Turnover by State and Territory, 2024



New South Wales had the highest turnover, at roughly \$135,639 million, equivalent to 31.1% of the national total. Victoria followed with around \$112,611 million or 25.8%, meaning these two states alone accounted for 56.8% of all turnover in 2024, a gap of roughly \$23,028 million between them despite only a 5.3 percentage point difference in share. Queensland held the third-largest share, with turnover of about \$89,987 million, representing 20.6% of the total, bringing the combined share of the three largest states to 77.4% and leaving the remaining five states and territories to account for just 22.6% between them.

Western Australia and South Australia contributed considerably smaller amounts, at around \$49,491 million (11.3%) and \$28,136 million (6.4%) respectively. Together with the top three states, these five regions accounted for over 95% of national turnover, leaving only a small remainder for the three smallest states and territories.

Tasmania, the Australian Capital Territory, and the Northern Territory recorded the lowest turnover figures, at about \$8,717 million, \$8,150 million and \$4,026 million respectively, equating to 2.0%, 1.9%, and 0.9% of the national total. Combined, these three regions contributed less than 5% of national turnover. The gap between the largest and smallest state is particularly pronounced, with New South Wales' turnover roughly 34 times that of the Northern Territory, a scale of difference that highlights how heavily concentrated Australian retail activity is in its most populous eastern states.

```
# Calculate annual total turnover for each state
annual_state_turnover <- turnover %>%
  filter(industry_group == "Total (Industry)", state != "Total (State)") %>%

group_by(year, state) %>%
  summarise( annual_turnover = sum( value, na.rm = TRUE),
             .groups = "drop")

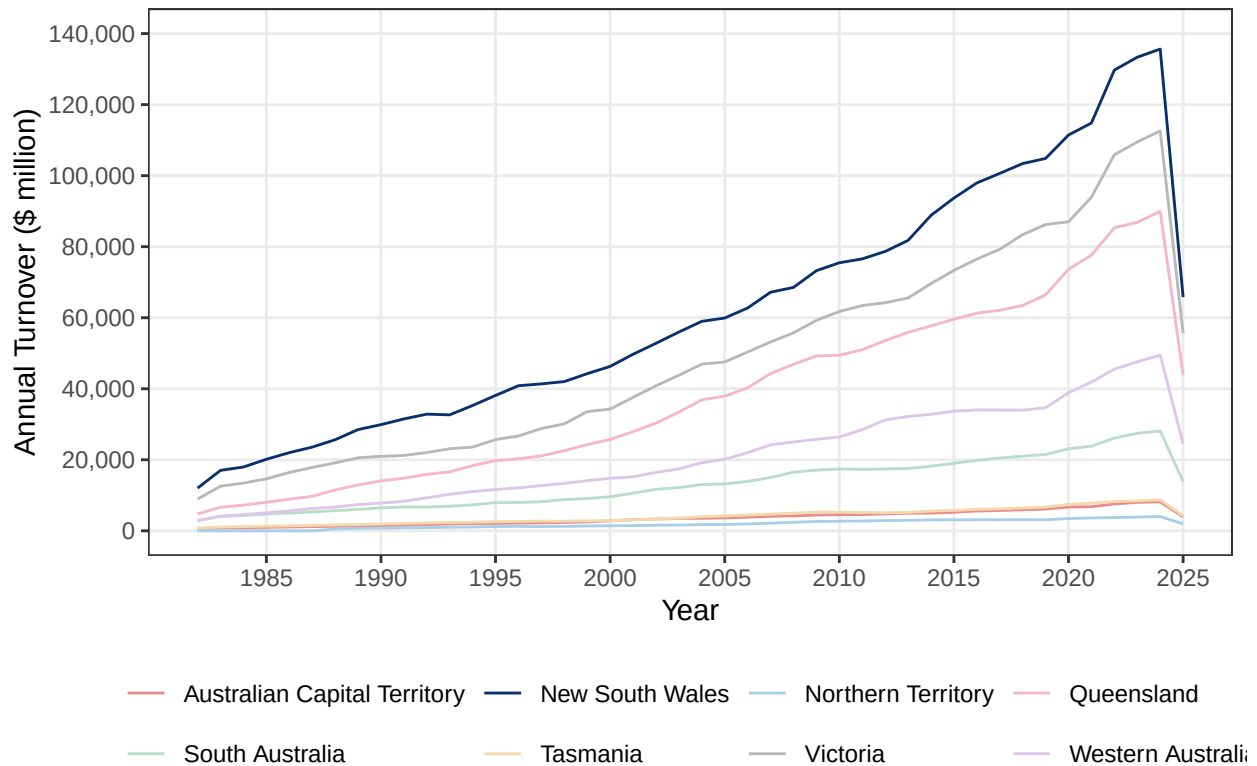
#plotting
```

```

ggplot( annual_state_turnover,
  aes(x = year,
      y = annual_turnover,
      colour = state,
      group = state)
) +
geom_line(linewidth = 0.5) +
scale_colour_manual(values = c("New South Wales" = "#08306B",
                                "Victoria" = "#B8B8B8",
                                "Queensland" = "#F5B6CD",
                                "Western Australia" = "#DFC5EB",
                                "South Australia" = "#B9DEC7",
                                "Tasmania" = "#F5D9AE",
                                "Australian Capital Territory" = "#E98B8B",
                                "Northern Territory" = "#A9CFE8")
) +
scale_x_continuous(breaks = seq(1985, 2025, by = 5) ) +
scale_y_continuous(breaks = seq(0,140000,by = 20000),
  labels = scales::label_comma(),
  limits = c(0, 140000)
) +
labs(title = "Annual Turnover by State and Territory",
  x = "Year",
  y = "Annual Turnover ($ million)",
  colour = NULL,
) +
theme_bw(base_size = 11) +
theme(plot.title = element_text(face = "bold",size = 15,hjust = 0.5),
  panel.grid.minor = element_blank(),
  legend.position = "bottom",
  legend.text = element_text(size = 9)
) +
guides(colour = guide_legend( nrow = 2, byrow = TRUE))

```

Annual Turnover by State and Territory



The line graph illustrates that New South Wales recorded the highest annual turnover of all states and territories in every year from the early 1980s through to 2025. Its line remains above Victoria's, the second-highest state, throughout the entire series, confirming that NSW's lead was continuous rather than something it gained or lost at different points in time.

New South Wales' turnover rose from around \$12,000 million in the early 1980s to a peak of approximately \$136,000 million in 2024. Victoria followed a broadly similar upward path, growing from about \$8,000 million to a peak of around \$113,000 million, while Queensland rose from roughly \$5,000 million to approximately \$90,000 million over the same timeframe. These three states maintained a consistent order throughout, with the gap between them widening steadily as turnover increased, particularly from early 2000s onward.

Western Australia and South Australia grew more moderately, reaching peaks of around \$50,000 million and \$29,000 million by 2024, up from under \$5,000 million each at the start. Both states tracked closely together until the early 2000s, when Western Australia began pulling ahead, and the gap between them continued widening through to 2024.

Tasmania, the Australian Capital Territory, and the Northern Territory recorded far lower turnover throughout, staying below \$10,000 million even at their 2024 peaks. Tasmania and the Australian Capital Territory tracked each other closely for most of the period, with their lines running almost together, while the Northern Territory remained consistently the lowest of the three. Despite their small scale, all three still grew several times over from their starting levels in the early 1980s.

Every state and territory shows a steep decline in the final data point around 2025, with New South Wales falling from roughly \$136,000 million to about \$66,000 million. Given this sudden and uniform drop, it likely reflects a partial or incomplete final year of data rather than an actual collapse in retail turnover.

#NSW in 2007 were chosen

```

#filter data
nsw_industry <- turnover %>%
filter(state == "New South Wales",
       year == 2007,
       industry_group != "Total (Industry)"
) %>%

#calculate the median turnover for each industry to divide them into high, mid and low turnover level

group_by(industry_group) %>%
mutate(industry_median = median(value)) %>%
ungroup() %>%
mutate(
  turnover_level = case_when(
    industry_median >= 1000 ~ "High turnover",
    industry_median >= 300 ~ "Mid turnover",
    TRUE ~ "Low turnover"),
  turnover_level = factor( turnover_level, levels = c("High turnover","Mid turnover","Low turnover"))
)

#plotting
ggplot(nsw_industry,
       aes(x = industry_group, y = value)
) +
geom_violin(
  scale = "width",
  adjust = 2,
  fill = "white",
  colour = "black",
  linewidth = 0.5
) +
geom_boxplot(
  width = 0.2,
  fill = "orange",
  colour = "black",
  outlier.shape = NA,
  alpha = 0.6,
  linewidth = 0.5
) +
geom_point(
  position = position_jitter(seed = 1, width = 0.05),
  colour = "grey20",
  size = 3,
  alpha = 0.65
) +
facet_wrap(
  ~turnover_level,
  scales = "free",
  ncol = 1
) +
scale_y_continuous(labels = scales::label_comma())
) +
scale_x_discrete(

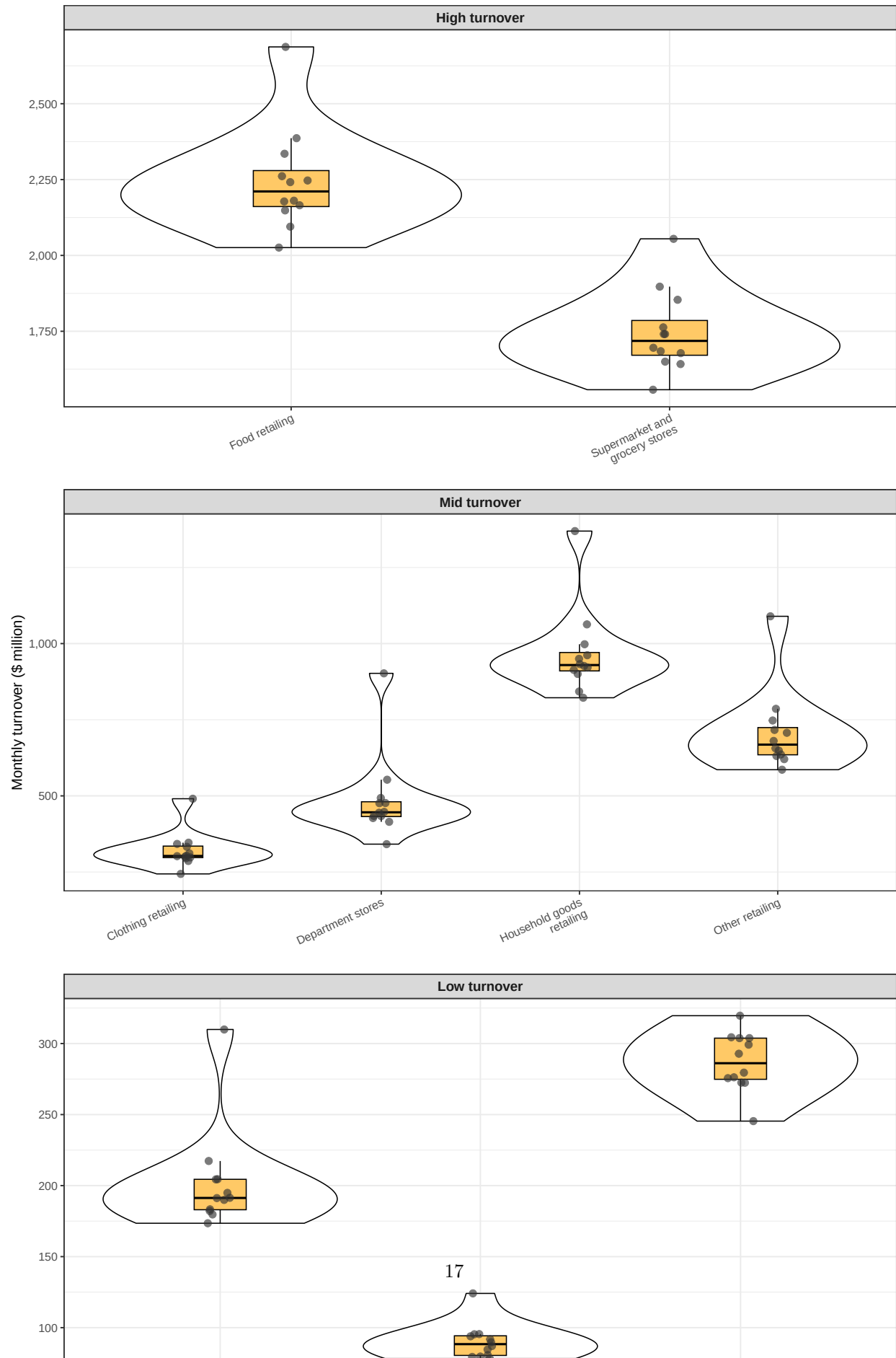
```

```

  labels = function(x) {stringr::str_wrap(x, width = 18)}
) +
labs(
  title = "NSW Turnover Distribution by Industry Group in 2007",
  x = "Industry Group",
  y = "Monthly turnover ($ million)"
) +
theme_bw(base_size = 14) +
theme(
  plot.title = element_text(face = "bold",size = 18, hjust = 0.5),
  strip.text = element_text(face = "bold", size = 13),
  axis.text.x = element_text( size = 11, angle = 25, hjust = 1),
  panel.spacing = unit(1.2, "lines")
)

```

NSW Turnover Distribution by Industry Group in 2007



====> SUGGEST SEASONAL PATTERN The industry's turnover values were divided into 3 tiers as below: - High turnover tier: Food retailing has the higher distribution, with a median turnover of about \$2,200 million and most values concentrated around \$2,150-\$2,300 million. Its violin narrows into a long upper tail, showing that only a small number of months recorded substantially higher turnover. Supermarket and grocery stores has a lower median of around \$1,700 million, with most values concentrated around \$1,600-\$1,800 million and a shorter upper tail. The boxes of these two industries do not overlap, indicating a clear difference in typical turnover between the two industries, rather than a difference caused only by one or two unusually high months.

- Middle turnover tier: The industries separate clearly by monthly turnover, with no overlap between their boxes. Household goods retailing ranks first with a median of about \$900 million and an upper tail reaching \$1,300 million. Other retailing ranks second with a median of around \$600 million and a longer relative upper tail reaching \$1,100 million. Department stores ranks third at approximately \$450 million, but it shows the strongest relative variability as most months are concentrated around \$400-\$500 million, while its upper tail reaches approximately \$900 million, more than twice its median. Clothing retailing ranks fourth with a median of \$300 million and has the narrowest box, with the middle 50% of observations concentrated around \$290-\$330 million, making it the most consistent industry during most months despite one higher observation of approximately \$490 million.
- Low turnover tier: Takeaway food services ranks first with a median monthly turnover of approximately \$280 million and has the most concentrated distribution in this tier. However, its box of approximately \$275-\$300 million overlaps slightly with clothing retailing's box of \$290-\$330 million. This overlap shows that these two industries had partly similar turnover despite being placed in different tiers. Liquor retailing ranks second with a median of \$190 million, but its long upper tail reaches up to \$310 million, meaning that at least one high month reached takeaway's range. Newspaper and book retailing remains clearly lowest, with a median of \$85 million and a maximum of \$125 million, below liquor retailing's minimum of \$175 million.

What happened to industry groups before, during and after the COVID-19 pandemic?

```
#calculate annual national turnover for each industry from 2017 to 2024
covid_annual <- turnover %>%
filter(
  state == "Total (State)",
  industry_group != "Total (Industry)",
  year >= 2017,
  year <= 2024
) %>%
group_by(industry_group, year) %>%
summarise(
  reported_months = sum(!is.na(value)),
  annual_turnover = sum(value, na.rm = TRUE),
  .groups = "drop")

#Rank actual turnover within each year
covid_annual %>%
arrange(year, desc(annual_turnover)) %>%
print(n = 72)
```

```
## # A tibble: 72 x 4
##   industry_group      year reported_months annual_turnover
##   <chr>             <dbl>         <int>         <dbl>
## 1 Food retailing    2017             12      125826.
## 2 Supermarket and grocery stores 2017             12      105226.
```

## 3 Household goods retailing	2017	12	54166.
## 4 Other retailing	2017	12	44463.
## 5 Department stores	2017	12	18650.
## 6 Takeaway food services	2017	12	18285.
## 7 Clothing retailing	2017	12	16214.
## 8 Liquor retailing	2017	12	11775.
## 9 Newspaper and book retailing	2017	12	3156.
## 10 Food retailing	2018	12	130551.
## 11 Supermarket and grocery stores	2018	12	109147.
## 12 Household goods retailing	2018	12	55058.
## 13 Other retailing	2018	12	45891.
## 14 Department stores	2018	12	18689.
## 15 Takeaway food services	2018	12	18687.
## 16 Clothing retailing	2018	12	16920.
## 17 Liquor retailing	2018	12	12139.
## 18 Newspaper and book retailing	2018	12	2914.
## 19 Food retailing	2019	12	135269.
## 20 Supermarket and grocery stores	2019	12	113414.
## 21 Household goods retailing	2019	12	55281.
## 22 Other retailing	2019	12	47401.
## 23 Takeaway food services	2019	12	19186.
## 24 Department stores	2019	12	18833.
## 25 Clothing retailing	2019	12	17452.
## 26 Liquor retailing	2019	12	12446.
## 27 Newspaper and book retailing	2019	12	2677.
## 28 Food retailing	2020	12	151546.
## 29 Supermarket and grocery stores	2020	12	124709.
## 30 Household goods retailing	2020	12	64958.
## 31 Other retailing	2020	12	52152.
## 32 Takeaway food services	2020	12	19065.
## 33 Department stores	2020	12	19042.
## 34 Liquor retailing	2020	12	16391.
## 35 Clothing retailing	2020	12	16291.
## 36 Newspaper and book retailing	2020	12	2371.
## 37 Food retailing	2021	12	152866.
## 38 Supermarket and grocery stores	2021	12	125012.
## 39 Household goods retailing	2021	12	67432.
## 40 Other retailing	2021	12	57263.
## 41 Takeaway food services	2021	12	20143.
## 42 Department stores	2021	12	18920.
## 43 Clothing retailing	2021	12	18697.
## 44 Liquor retailing	2021	12	17079.
## 45 Newspaper and book retailing	2021	12	2301.
## 46 Food retailing	2022	12	160937.
## 47 Supermarket and grocery stores	2022	12	132009.
## 48 Household goods retailing	2022	12	72165.
## 49 Other retailing	2022	12	64160.
## 50 Clothing retailing	2022	12	23265.
## 51 Takeaway food services	2022	12	22149.
## 52 Department stores	2022	12	21950.
## 53 Liquor retailing	2022	12	17720.
## 54 Newspaper and book retailing	2022	12	2190.
## 55 Food retailing	2023	12	168449.
## 56 Supermarket and grocery stores	2023	12	139197.

## 57 Household goods retailing	2023	12	69536
## 58 Other retailing	2023	12	64758.
## 59 Clothing retailing	2023	12	23952.
## 60 Takeaway food services	2023	12	23854.
## 61 Department stores	2023	12	22687.
## 62 Liquor retailing	2023	12	17967.
## 63 Newspaper and book retailing	2023	12	1995.
## 64 Food retailing	2024	12	173604.
## 65 Supermarket and grocery stores	2024	12	143396
## 66 Household goods retailing	2024	12	70416
## 67 Other retailing	2024	12	68326.
## 68 Takeaway food services	2024	12	24530
## 69 Clothing retailing	2024	12	24415.
## 70 Department stores	2024	12	22854.
## 71 Liquor retailing	2024	12	17981.
## 72 Newspaper and book retailing	2024	12	2014.

```

#use each industry's 2019 turnover as the baseline
baseline_2019 <- covid_annual %>%
filter(year == 2019) %>%
select(industry_group, turnover_2019 = annual_turnover)
baseline_2019 %>%
arrange(desc(turnover_2019))

```

```

## # A tibble: 9 x 2
##   industry_group      turnover_2019
##   <chr>              <dbl>
## 1 Food retailing      135269.
## 2 Supermarket and grocery stores 113414.
## 3 Household goods retailing      55281
## 4 Other retailing      47401.
## 5 Takeaway food services      19186.
## 6 Department stores      18833.
## 7 Clothing retailing      17452.
## 8 Liquor retailing      12446
## 9 Newspaper and book retailing      2677.

```

```

# Convert annual turnover into an index where 2019 equals 100
covid_annual <- covid_annual %>%
left_join(baseline_2019, by = "industry_group") %>%
mutate(
  turnover_index = annual_turnover / turnover_2019 * 100,
  period = case_when(
    year <= 2019 ~ "Before COVID-19",
    year <= 2021 ~ "During COVID-19",
    TRUE       ~ "After COVID-19"),
  period = factor(
    period,
    levels = c("Before COVID-19", "During COVID-19", "After COVID-19"))
)
#check annual turnover, index and period for every industry
covid_annual %>%
select(

```

```

industry_group,
year,
annual_turnover,
turnover_2019,
turnover_index,
period
) %>%
arrange(industry_group, year) %>%
print(n = 72)

```

```

## # A tibble: 72 x 6
##   industry_group      year annual_turnover turnover_2019 turnover_index period
##   <chr>          <dbl>         <dbl>         <dbl>         <dbl> <fct>
## 1 Clothing retailing  2017         16214.         17452.          92.9 Befor~
## 2 Clothing retailing  2018         16920          17452.          97.0 Befor~
## 3 Clothing retailing  2019         17452.         17452.         100 Befor~
## 4 Clothing retailing  2020         16291.         17452.          93.3 Durin~
## 5 Clothing retailing  2021         18697          17452.         107. Durin~
## 6 Clothing retailing  2022         23265          17452.         133. After~
## 7 Clothing retailing  2023         23952.         17452.         137. After~
## 8 Clothing retailing  2024         24415.         17452.         140. After~
## 9 Department stores  2017         18650.         18833.          99.0 Befor~
## 10 Department stores  2018         18689.         18833.          99.2 Befor~
## 11 Department stores  2019         18833.         18833.         100 Befor~
## 12 Department stores  2020         19042.         18833.         101. Durin~
## 13 Department stores  2021         18920.         18833.         100. Durin~
## 14 Department stores  2022         21950.         18833.         117. After~
## 15 Department stores  2023         22687.         18833.         120. After~
## 16 Department stores  2024         22854.         18833.         121. After~
## 17 Food retailing     2017        125826.        135269.          93.0 Befor~
## 18 Food retailing     2018        130551.        135269.          96.5 Befor~
## 19 Food retailing     2019        135269.        135269.         100 Befor~
## 20 Food retailing     2020        151546.        135269.         112. Durin~
## 21 Food retailing     2021        152866.        135269.         113. Durin~
## 22 Food retailing     2022        160937.        135269.         119. After~
## 23 Food retailing     2023        168449.        135269.         125. After~
## 24 Food retailing     2024        173604.        135269.         128. After~
## 25 Household goods re~ 2017         54166.         55281          98.0 Befor~
## 26 Household goods re~ 2018         55058.         55281          99.6 Befor~
## 27 Household goods re~ 2019         55281          55281         100 Befor~
## 28 Household goods re~ 2020         64958.         55281         118. Durin~
## 29 Household goods re~ 2021         67432.         55281         122. Durin~
## 30 Household goods re~ 2022         72165          55281         131. After~
## 31 Household goods re~ 2023         69536          55281         126. After~
## 32 Household goods re~ 2024         70416          55281         127. After~
## 33 Liquor retailing   2017         11775.         12446          94.6 Befor~
## 34 Liquor retailing   2018         12139.         12446          97.5 Befor~
## 35 Liquor retailing   2019         12446          12446         100 Befor~
## 36 Liquor retailing   2020         16391.         12446         132. Durin~
## 37 Liquor retailing   2021         17079          12446         137. Durin~
## 38 Liquor retailing   2022         17720.         12446         142. After~
## 39 Liquor retailing   2023         17967.         12446         144. After~
## 40 Liquor retailing   2024         17981.         12446         144. After~

```

## 41 Newspaper and book~	2017	3156.	2677.	118.	Befor~
## 42 Newspaper and book~	2018	2914.	2677.	109.	Befor~
## 43 Newspaper and book~	2019	2677.	2677.	100	Befor~
## 44 Newspaper and book~	2020	2371.	2677.	88.5	Durin~
## 45 Newspaper and book~	2021	2301.	2677.	85.9	Durin~
## 46 Newspaper and book~	2022	2190	2677.	81.8	After~
## 47 Newspaper and book~	2023	1995.	2677.	74.5	After~
## 48 Newspaper and book~	2024	2014.	2677.	75.2	After~
## 49 Other retailing	2017	44463.	47401.	93.8	Befor~
## 50 Other retailing	2018	45891.	47401.	96.8	Befor~
## 51 Other retailing	2019	47401.	47401.	100	Befor~
## 52 Other retailing	2020	52152.	47401.	110.	Durin~
## 53 Other retailing	2021	57263	47401.	121.	Durin~
## 54 Other retailing	2022	64160.	47401.	135.	After~
## 55 Other retailing	2023	64758.	47401.	137.	After~
## 56 Other retailing	2024	68326.	47401.	144.	After~
## 57 Supermarket and gr~	2017	105226.	113414.	92.8	Befor~
## 58 Supermarket and gr~	2018	109147.	113414.	96.2	Befor~
## 59 Supermarket and gr~	2019	113414.	113414.	100	Befor~
## 60 Supermarket and gr~	2020	124709.	113414.	110.	Durin~
## 61 Supermarket and gr~	2021	125012.	113414.	110.	Durin~
## 62 Supermarket and gr~	2022	132009	113414.	116.	After~
## 63 Supermarket and gr~	2023	139197.	113414.	123.	After~
## 64 Supermarket and gr~	2024	143396	113414.	126.	After~
## 65 Takeaway food serv~	2017	18285.	19186.	95.3	Befor~
## 66 Takeaway food serv~	2018	18687.	19186.	97.4	Befor~
## 67 Takeaway food serv~	2019	19186.	19186.	100	Befor~
## 68 Takeaway food serv~	2020	19065.	19186.	99.4	Durin~
## 69 Takeaway food serv~	2021	20143.	19186.	105.	Durin~
## 70 Takeaway food serv~	2022	22149.	19186.	115.	After~
## 71 Takeaway food serv~	2023	23854.	19186.	124.	After~
## 72 Takeaway food serv~	2024	24530	19186.	128.	After~

```

#Define the beginning and end of each period for graph shading
covid_periods <- tibble(
  period = factor(
    c("Before COVID-19", "During COVID-19", "After COVID-19"),
    levels = c("Before COVID-19", "During COVID-19", "After COVID-19")),
  xmin = c(2016.5, 2019.5, 2021.5),
  xmax = c(2019.5, 2021.5, 2024.5)
)

covid_periods

```

```

## # A tibble: 3 x 3
##   period      xmin  xmax
##   <fct>      <dbl> <dbl>
## 1 Before COVID-19 2016. 2020.
## 2 During COVID-19 2020. 2022.
## 3 After COVID-19  2022. 2024.

```

```

# Does New South Wales's overall leadership mean that it leads every individual industry?

```

```

# Calculate and rank total state turnover in 2024
overall_state_2024 <- turnover %>%
filter(
  year == 2024,
  industry_group == "Total (Industry)",
  state != "Total (State)"
) %>%
group_by(state) %>%
summarise(
  reported_months = sum(!is.na(value)),
  annual_turnover = sum(value, na.rm = TRUE),
  .groups = "drop"
) %>%
arrange(desc(annual_turnover))

overall_state_2024

```

```

## # A tibble: 8 x 3
##   state                reported_months annual_turnover
##   <chr>                  <int>          <dbl>
## 1 New South Wales         12        135639.
## 2 Victoria                12        112611
## 3 Queensland              12         89987.
## 4 Western Australia       12         49491.
## 5 South Australia         12         28136.
## 6 Tasmania                12          8717.
## 7 Australian Capital Territory 12          8150.
## 8 Northern Territory       12          4026.

```

```

#Calculate each state's annual turnover within each industry
state_industry_2024 <- turnover %>%
filter(
  year == 2024,
  state != "Total (State)",
  industry_group != "Total (Industry)"
) %>%
group_by(industry_group, state) %>%
summarise(
  reported_months = sum(!is.na(value)),
  annual_turnover = sum(value, na.rm = TRUE),
  .groups = "drop"
) %>%
group_by(industry_group) %>%
mutate(
  industry_total = sum(annual_turnover),
  turnover_share = annual_turnover / industry_total * 100,
  industry_rank = min_rank(desc(annual_turnover))
) %>%
ungroup()

# Calculate each state's share and rank within its industry
state_industry_2024 <- state_industry_2024 %>%
mutate(

```

```

state_short = recode(state,
  "New South Wales" = "NSW",
  "Victoria" = "VIC",
  "Queensland" = "QLD",
  "Western Australia" = "WA",
  "South Australia" = "SA",
  "Tasmania" = "TAS",
  "Australian Capital Territory" = "ACT",
  "Northern Territory" = "NT"),
state_short = factor(state_short, levels = c("NSW", "VIC", "QLD", "WA", "SA", "TAS", "ACT", "NT")),
industry_group = forcats::fct_reorder(industry_group, industry_total, .desc = TRUE),
percentage_label = paste0(round(turnover_share, 1), "%"),
label_colour = if_else(turnover_share >= 24, "white", "black"),
label_style = if_else(industry_rank == 1, "bold", "plain")
)

# check the three leading states in every industry
state_industry_2024 %>%
filter(industry_rank <= 3) %>%
arrange(industry_group, industry_rank) %>%
select(
  industry_group,
  state,
  annual_turnover,
  turnover_share,
  industry_rank
) %>%
print(n = 27)

```

```

## # A tibble: 27 x 5
##   industry_group      state annual_turnover turnover_share industry_rank
##   <fct>             <chr>         <dbl>         <dbl>         <int>
## 1 Food retailing    New ~           52805.         30.4           1
## 2 Food retailing    Vict~           42537.         24.5           2
## 3 Food retailing    Quee~           37244.         21.5           3
## 4 Supermarket and grocery s~ New ~           43086.         30.0           1
## 5 Supermarket and grocery s~ Vict~           35336.         24.6           2
## 6 Supermarket and grocery s~ Quee~           30992.         21.6           3
## 7 Household goods retailing New ~           21504.         30.5           1
## 8 Household goods retailing Vict~           18649.         26.5           2
## 9 Household goods retailing Quee~           14182.         20.1           3
## 10 Other retailing   New ~           20505.         30.0           1
## 11 Other retailing   Vict~           18994.         27.8           2
## 12 Other retailing   Quee~           13559.         19.8           3
## 13 Takeaway food services New ~           6925.         28.2           1
## 14 Takeaway food services Vict~           5655.         23.1           2
## 15 Takeaway food services Quee~           5579.         22.7           3
## 16 Clothing retailing New ~           9603.         39.3           1
## 17 Clothing retailing Vict~           6925.         28.4           2
## 18 Clothing retailing Quee~           4384.         18.0           3
## 19 Department stores New ~           7017.         31.6           1
## 20 Department stores Vict~           5489.         24.7           2
## 21 Department stores Quee~           4973.         22.4           3

```

## 22 Liquor retailing	New ~	5399.	41.0	1
## 23 Liquor retailing	Vict~	4351.	33.1	2
## 24 Liquor retailing	West~	2164.	16.4	3
## 25 Newspaper and book retail~	Quee~	598.	29.7	1
## 26 Newspaper and book retail~	Vict~	498.	24.7	2
## 27 Newspaper and book retail~	New ~	465.	23.1	3

Is Victoria catching up with New South Wales in retail turnover?

#Calculate annual state turnover for full years of 1989-2024

```
annual_state_turnover <- turnover %>%
filter(
  year >= 1989,
  year <= 2024,
  industry_group == "Total (Industry)",
  state != "Total (State)"
) %>%
group_by(year, state) %>%
summarise(
  reported_months = sum(!is.na(value)),
  annual_turnover = sum(value, na.rm = TRUE),
  .groups = "drop"
)
```

Rank all states from highest to lowest turnover within each year

```
ranked_state_turnover <- annual_state_turnover %>%
group_by(year) %>%
arrange(
  desc(annual_turnover),
  .by_group = TRUE
) %>%
mutate(
  annual_rank = row_number()
) %>%
ungroup()
```

#Find the highest-turnover state in each year

```
annual_first_states <- ranked_state_turnover %>%
filter(annual_rank == 1) %>%
select(year,
  first_state = state,
  first_state_turnover = annual_turnover)
```

#Find the second-highest state in each year

```
annual_second_states <- ranked_state_turnover %>%
filter(annual_rank == 2) %>%
select(
  year,
  second_state = state,
  second_state_turnover = annual_turnover
)
```

#Combine the first and second-ranked states and calculate the difference between them

```
annual_state_gap <- annual_first_states %>%
```

```

left_join(annual_second_states,by = "year") %>%
mutate( dollar_gap = first_state_turnover - second_state_turnover,
        percentage_gap = dollar_gap / second_state_turnover * 100)

#year with the largest relative gap
annual_state_gap %>%
slice_max(percentage_gap,n = 1)

```

```

## # A tibble: 1 x 7
##   year first_state first_state_turnover second_state second_state_turnover
##   <dbl> <chr>          <dbl> <chr>          <dbl>
## 1  1996 New South Wales      40853. Victoria      26701.
## # i 2 more variables: dollar_gap <dbl>, percentage_gap <dbl>

```

```

#year with the smallest relative gap
annual_state_gap %>%
slice_min(percentage_gap, n = 1)

```

```

## # A tibble: 1 x 7
##   year first_state first_state_turnover second_state second_state_turnover
##   <dbl> <chr>          <dbl> <chr>          <dbl>
## 1  2024 New South Wales      135639. Victoria      112611
## # i 2 more variables: dollar_gap <dbl>, percentage_gap <dbl>

```

```

# Which industries show the strongest seasonal variation?

```

```

# Calculate monthly turnover relative to each industry's average month within the same year

```

```

monthly_seasonality <- turnover %>%
filter(
  year >= 2015,
  year <= 2024,
  state == "Total (State)",
  industry_group != "Total (Industry)"
) %>%
group_by(industry_group, year) %>%
mutate(
  yearly_average = mean(value, na.rm = TRUE),
  monthly_index = value / yearly_average * 100
) %>%
ungroup() %>%
group_by(industry_group, month) %>%
summarise(
  average_monthly_index = mean(monthly_index,na.rm = TRUE),
  .groups = "drop"
) %>%

```

```

# Convert month numbers into abbreviated month names
mutate(month_name = month.abb[month])

```

```

# Identify each industry's lowest and highest months and calculate its seasonal range
seasonal_summary <- monthly_seasonality %>%
group_by(industry_group) %>%

```

```

summarise(
  lowest_month = month_name[which.min(average_monthly_index)],
  lowest_index = min(average_monthly_index),
  highest_month = month_name[which.max(average_monthly_index)],
  highest_index = max(average_monthly_index),
  seasonal_range = highest_index - lowest_index,
  .groups = "drop"
) %>%
arrange(desc(seasonal_range))

industry_order <- seasonal_summary %>%
  arrange(seasonal_range) %>%
  pull(industry_group)

#ensure the same industry order to the monthly observations
monthly_seasonality <- monthly_seasonality %>%
  mutate(
    industry_group = factor(
      industry_group,
      levels = industry_order
    )
  )

#ensure the same order to the seasonal summary
seasonal_summary <- seasonal_summary %>%
  mutate(
    industry_group = factor(
      industry_group,
      levels = industry_order
    )
  )

# check industries ranked by seasonal variation
seasonal_summary

```

```

## # A tibble: 9 x 6
##   industry_group      lowest_month lowest_index highest_month highest_index
##   <fct>              <chr>          <dbl> <chr>          <dbl>
## 1 Department stores    Feb              72.0 Dec              178.
## 2 Clothing retailing   Feb              76.9 Dec              158.
## 3 Liquor retailing     Feb              84.6 Dec              156.
## 4 Newspaper and book reta~ Jun              90.8 Dec              140.
## 5 Other retailing      Feb              86.9 Dec              133.
## 6 Household goods retaili~ Feb              86.7 Dec              129.
## 7 Food retailing       Feb              90.9 Dec              119.
## 8 Takeaway food services Feb              88.0 Dec              114.
## 9 Supermarket and grocery~ Feb              91.7 Dec              114.
## # i 1 more variable: seasonal_range <dbl>

```

b)

```

#covid plotting
ggplot(
  covid_annual,

```

```

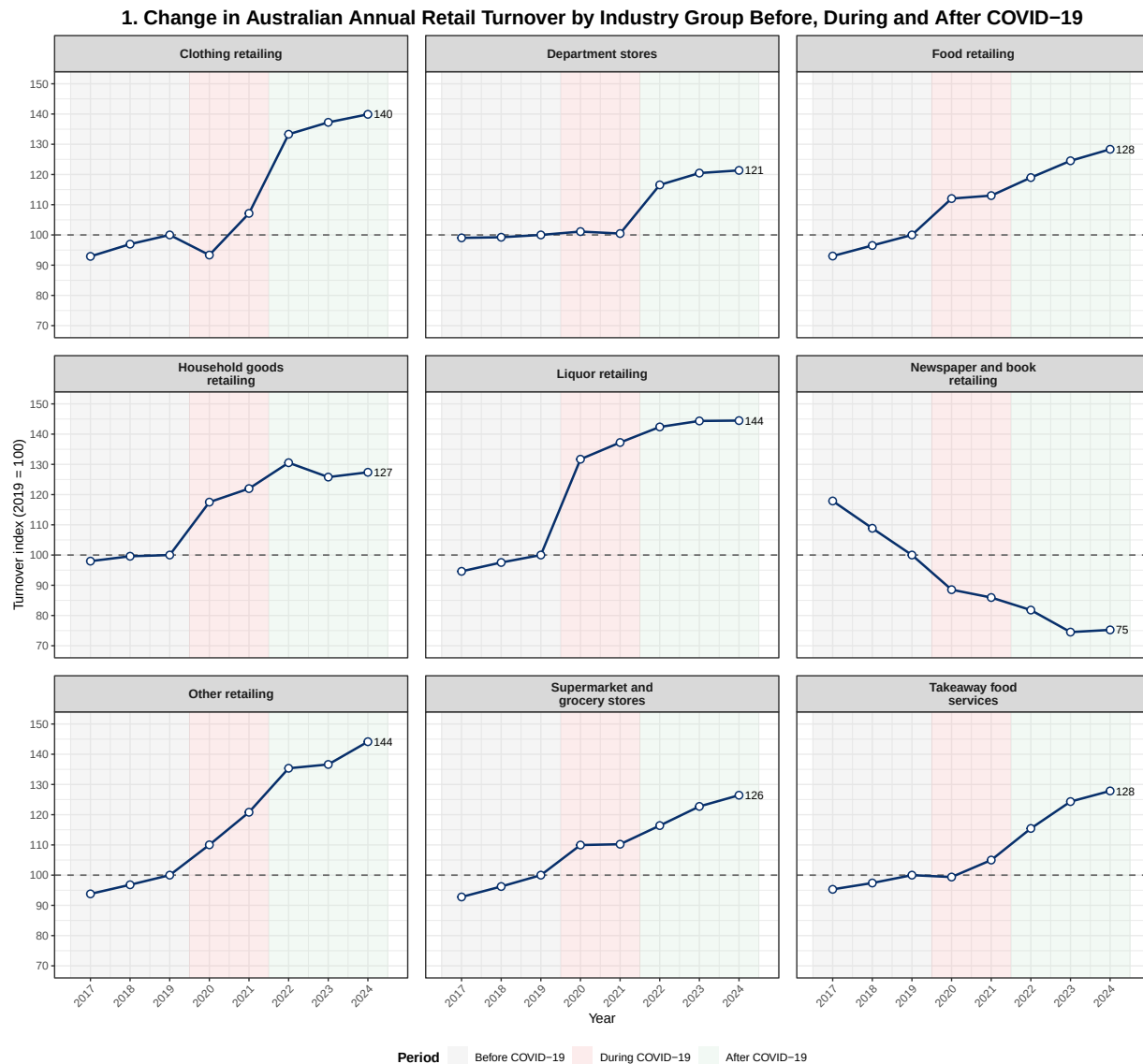
aes(
  x = year,
  y = turnover_index,
  group = industry_group)
) +
geom_rect(
  data = covid_periods,
  aes(
    xmin = xmin,
    xmax = xmax,
    ymin = -Inf,
    ymax = Inf,
    fill = period),
  inherit.aes = FALSE,
  alpha = 0.25
) +
geom_hline(
  yintercept = 100,
  linetype = "dashed",
  colour = "grey40",
  linewidth = 0.6
) +
geom_line(colour = "#08306B", linewidth = 0.9
) +
geom_point(
  shape = 21,
  fill = "white",
  colour = "#08306B",
  size = 2.3,
  stroke = 0.8
) +
geom_text(
  data = covid_annual %>%
  filter(year == 2024),
  aes(label = round(turnover_index)),
  nudge_x = 0.15,
  hjust = 0,
  size = 3
) +
facet_wrap(
  ~industry_group,
  ncol = 3,
  labeller = label_wrap_gen(width = 22)
) +
scale_x_continuous(breaks = 2017:2024, limits = c(2016.5, 2024.6))
) +
scale_y_continuous(breaks = seq(70, 150, by = 10))
) +
scale_fill_manual(
  values = c("Before COVID-19" = "#D9D9D9", "During COVID-19" = "#F4B6B6", "After COVID-19" = "#CFE8D5")
) +
coord_cartesian(ylim = c(70, 150), clip = "off")
) +

```

```

labs(
  title = "1. Change in Australian Annual Retail Turnover by Industry Group Before, During and After COVID-19",
  x = "Year",
  y = "Turnover index (2019 = 100)",
  fill = "Period"
) +
theme_bw(base_size = 11) +
theme(plot.title = element_text(face = "bold",size = 16,hjust=0.5),
strip.text = element_text(face = "bold",size = 10),
axis.text.x = element_text(angle = 45,hjust = 1),
legend.position = "bottom",
legend.title = element_text(face = "bold"),
panel.spacing = unit(1, "lines"),
plot.margin = margin(5.5, 20, 5.5, 5.5)
)

```



- What happened to the industry group before, during and after Covid pandemic?

The indexed line charts show that Australian industries followed different trends before, during and after the COVID-19 pandemic. Indexing each industry to its 2019 turnover (2019 = 100) makes relative change directly comparable across industries. For example, an index of 132 means turnover was 32% above that industry's own 2019 level.

Regarding the charts, it is noticeable that before COVID-19, most industries were broadly flat to mildly increasing as they approached the 2019 baseline. Clothing, food retailing and supermarket and grocery rose from around 92 to 100. Department stores were essentially flat across the period around 98 to 100. The clear exception was newspaper and book retailing, which declined from around 119 in 2017 to 100 in 2019, highlighting its downward trend was already happened before the pandemic and cannot be attributed to COVID-19.

During covid period, there are two separate groups of industries. Categories related to at-home consumption and essential spending rose immediately and continued climbing through 2021. Food retailing grew from 100 to roughly 113, supermarkets and grocery stores to around 110, household goods rose to approximately 121 and other retailing reached 120 in 2021, likely because people spent more on their homes while staying indoors during lock downs. Liquor retailing rose the most of any industry, reaching about 131 in 2020 and about 136 in 2021.

The second group stayed flat or even fell. Clothing dropped to about 93 in 2020 and only partly recovered to about 107 in 2021, still below its 2019 level for almost the whole period. Newspaper and book retailing kept falling, reaching about 88 in 2020 and about 86 in 2021. Department stores and takeaway food services showed no growth, staying close to 100 throughout the period.

After COVID-19 (2022 onward), growth speeds up across almost every industry. Clothing shows the strongest recovery, moving from about 107 at the end of the COVID period to 133 in 2022, 137 in 2023 and 140 in 2024. This is the clearest case of an industry that fell during COVID and then recovered strongly afterward. Department stores and takeaway food services also start growing again after staying almost flat during COVID. Other retailing follows a similar path and also reaches 144. Noticeably, while Liquor retailing keeps growing but at a much slower pace than during 2020–2021, Supermarkets and grocery stores observed a faster growing speed, reaching 126 by 2024. Food retailing continues steady, more moderate growth into the high 120s.

Household goods retailing and newspaper and book retailing are the two exceptions to this recovery pattern. Household goods peaked at about 131 in 2022, then pulled back to 125 in 2023 before growing up slightly to 127 in 2024, its only pullback in the entire dataset after an initial post-COVID recovery. Newspaper and book retailing keeps declining through the entire post-COVID period, reaching 75 by 2024. It is the only industry that finishes below its 2019 baseline and by a large margin.

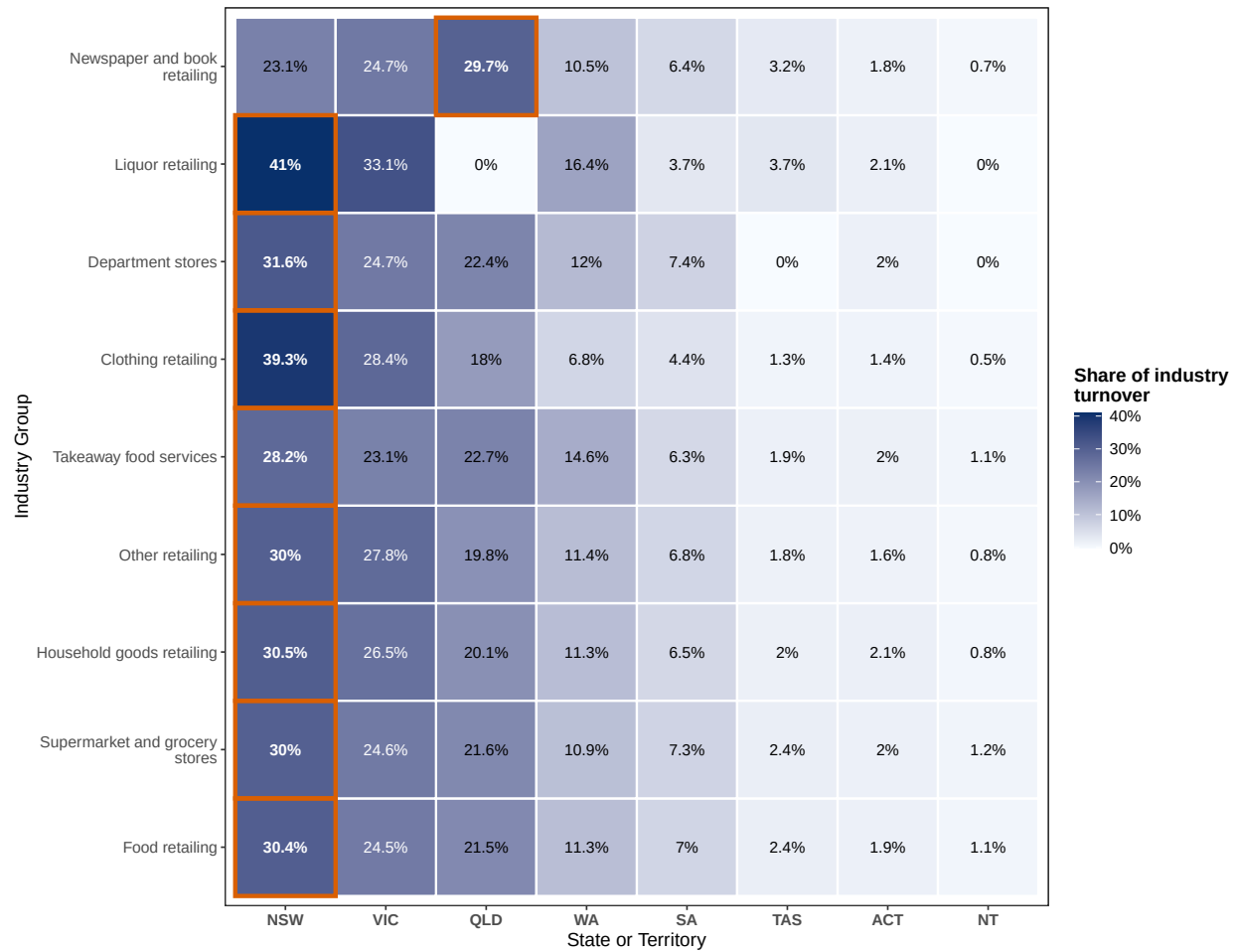
```
ggplot(
  state_industry_2024,
  aes(
    x = state_short,
    y = industry_group,
    fill = turnover_share)
) +
geom_tile(
  colour = "white",
  linewidth = 0.7
) +
geom_tile(
  data = state_industry_2024 %>%
  filter(industry_rank == 1),
  fill = NA,
  colour = "#D95F02",
  linewidth = 1.3
)
```

```

) +
geom_text(
  aes(label = percentage_label,
      colour = label_colour,
      fontface = label_style),
  size = 3.5
) +
scale_fill_gradient(
  low = "#F7FBFF",
  high = "#08306B",
  labels = scales::label_number(accuracy = 1, suffix = "%")
) +
scale_colour_identity() +
scale_y_discrete(labels = function(x) {stringr::str_wrap(x, width = 27)})
) +
labs(title = "2. State and Territory Shares of Turnover by Industry",
     x = "State or Territory",
     y = "Industry Group",
     fill = "Share of industry\nturnover",
     caption = paste("Each row sums to 100% across the eight states and territories.")
) +
theme_bw() +
theme(
  text = element_text(size = 12),
  plot.title = element_text(face = "bold", size = 16, hjust=0.5),
  axis.text.x = element_text(face = "bold"),
  panel.grid = element_blank(),
  legend.title = element_text(face = "bold")
)

```

2. State and Territory Shares of Turnover by Industry



- Does New South Wales highest turnover over time also mean it leads every industry?

New South Wales held the largest share in liquor retailing, at 41%, followed by clothing retailing at 39.3%, both considerably higher than Victoria's corresponding shares of 33.1% and 28.4%. NSW's first place seemed to be narrower in the remaining industries, ranging from 28.2% in takeaway food services to 31.6% in department stores, with Victoria consistently placed second in each case, at between 23.1% and 27.8%.

The sole exception was newspaper and book retailing, in which Queensland recorded the highest share at 29.7%, ahead of Victoria at 24.7% and New South Wales at 23.1%, the only instance in the chart where NSW was not the leading contributor.

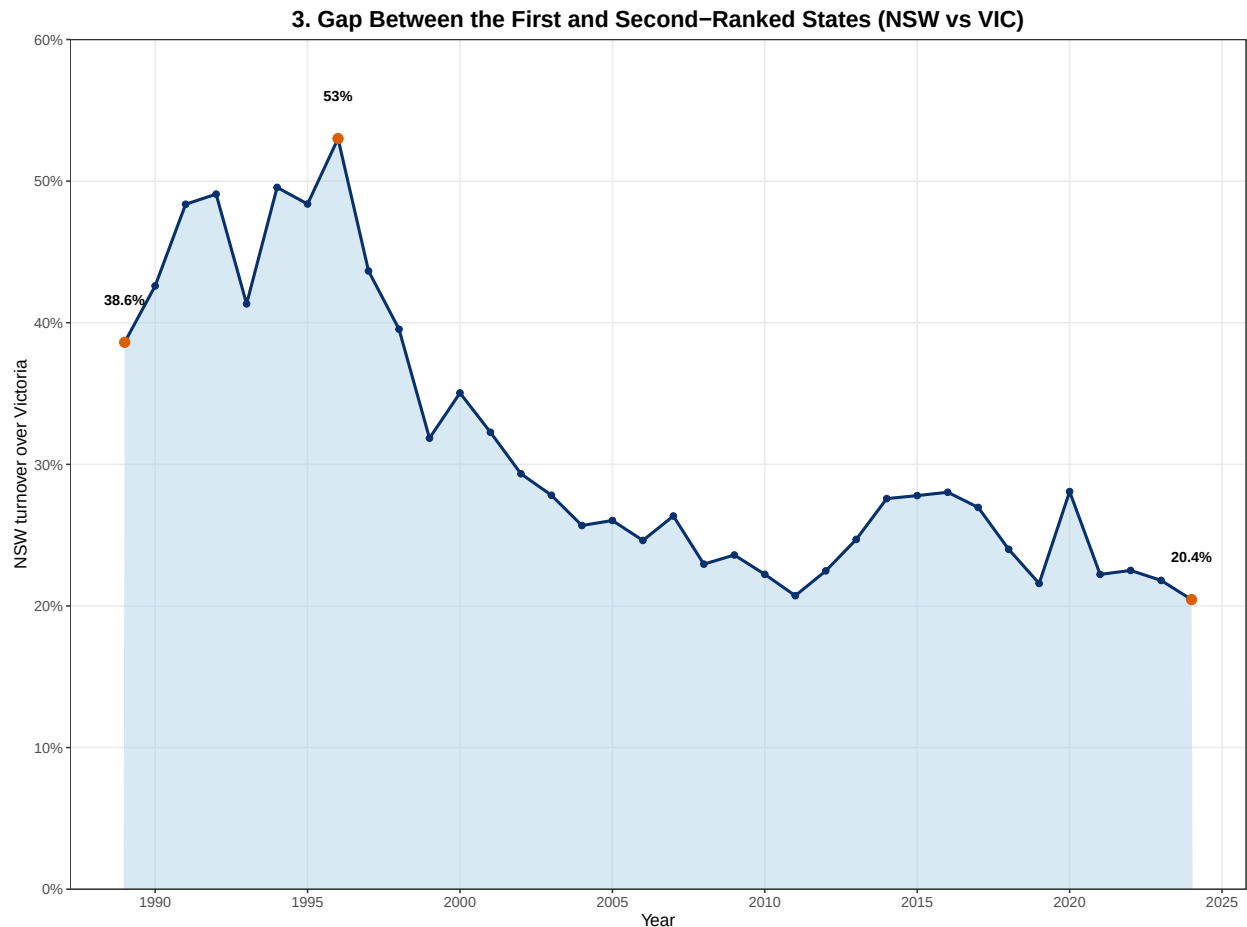
Western Australia consistently ranked fourth across all nine industries, with shares generally between 6.8% and 16.4%, while South Australia followed in fifth place, contributing between 3.7% and 7.4%. Tasmania, the ACT, and the Northern Territory contributed the smallest shares throughout, mostly remaining below 4%, and in three instances, liquor retailing in Queensland, department stores in Tasmania, and liquor retailing in the Northern Territory, turnover share fell to 0%.

```
ggplot(
  annual_state_gap,
  aes(
    x = year,
    y = percentage_gap
  )
)
```

```

) +
geom_area(
  fill = "#A9CFE8",
  alpha = 0.45
) +
geom_line(colour = "#08306B", linewidth = 1
) +
geom_point(colour = "#08306B", size = 1.8
) +
geom_point(
  data = annual_state_gap %>%
  filter(year %in% c(1989, 1996, 2024)),
  colour = "#D95F02",
  size = 3
) +
geom_text(
  data = annual_state_gap %>%
  filter(year %in% c(1989, 1996, 2024)),
  aes(
    label = paste0(round(percentage_gap, 1), "%")
  ),
  nudge_y = 3,
  fontface = "bold",
  size = 3.5
) +
scale_x_continuous(breaks = seq(1990, 2025, by = 5)
) +
scale_y_continuous(
  limits = c(0, 60),
  breaks = seq(0, 60, by = 10),
  labels = scales::label_number(suffix = "%"),
  expand = c(0, 0)
) +
labs(title = "3. Gap Between the First and Second-Ranked States (NSW vs VIC)",
     x = "Year",
     y = "NSW turnover over Victoria",
) +
theme_bw() +
theme(
  text = element_text(size = 12),
  plot.title = element_text(face = "bold", size = 16, hjust=0.5),
  panel.grid.minor = element_blank()
)

```



- Is Victoria Catching Up with New South Wales in Retail Turnover?

At the start of the period, in 1989, NSW's turnover exceeded Victoria's by 38.6%. This gap rose steadily over the following seven years, with only a brief dip around 1993, to reach a peak of 53% in 1996, the highest figure recorded across the entire timeframe. From this point, the gap declined sharply and almost continuously until 1999, falling to just under 32%, although there was a temporary rise to around 35% in 2000 before the downward trend resumed.

From the early 2000s onwards, the gap continued to narrow, though at a much slower and more irregular pace, moving broadly within a range of 21% to 28% for the next two decades. Two brief upward spikes interrupted this general decline, one around 2015 to 2016 and a more pronounced one in 2020, when the gap climbed back up to approximately 28%. Aside from these fluctuations, however, the overall trajectory from 2000 to 2024 remained downward, with the gap eventually falling to 20.4% by 2024, its lowest point in the entire 35-year period.

```
ggplot() +
  geom_vline(
    xintercept = 100,
    linetype = "dashed",
    colour = "grey50",
    linewidth = 0.6
  ) +
  geom_segment(
```

```

data = seasonal_summary,
aes(
  x = lowest_index,
  xend = highest_index,
  y = industry_group,
  yend = industry_group
),
colour = "grey70",
linewidth = 2
) +
geom_point(
  data = monthly_seasonality,
  aes(
    x = average_monthly_index,
    y = industry_group
  ),
  position = position_jitter(height = 0.08,width = 0, seed = 1
),
  colour = "grey35",
  size = 1.8,
  alpha = 0.55
) +
geom_point(
  data = seasonal_summary,
  aes(
    x = lowest_index,
    y = industry_group,
    colour = "Lowest month"
  ),
  size = 3.5
) +
geom_point(
  data = seasonal_summary,
  aes(
    x = highest_index,
    y = industry_group,
    colour = "Highest month"
  ),
  size = 3.5
) +
geom_text(
  data = seasonal_summary,
  aes(
    x = lowest_index,
    y = industry_group,
    label = paste0(
      lowest_month,
      " ",
      round(lowest_index, 1))
  ),
  hjust = 1.15,
  colour = "#08306B",
  size = 3.2

```

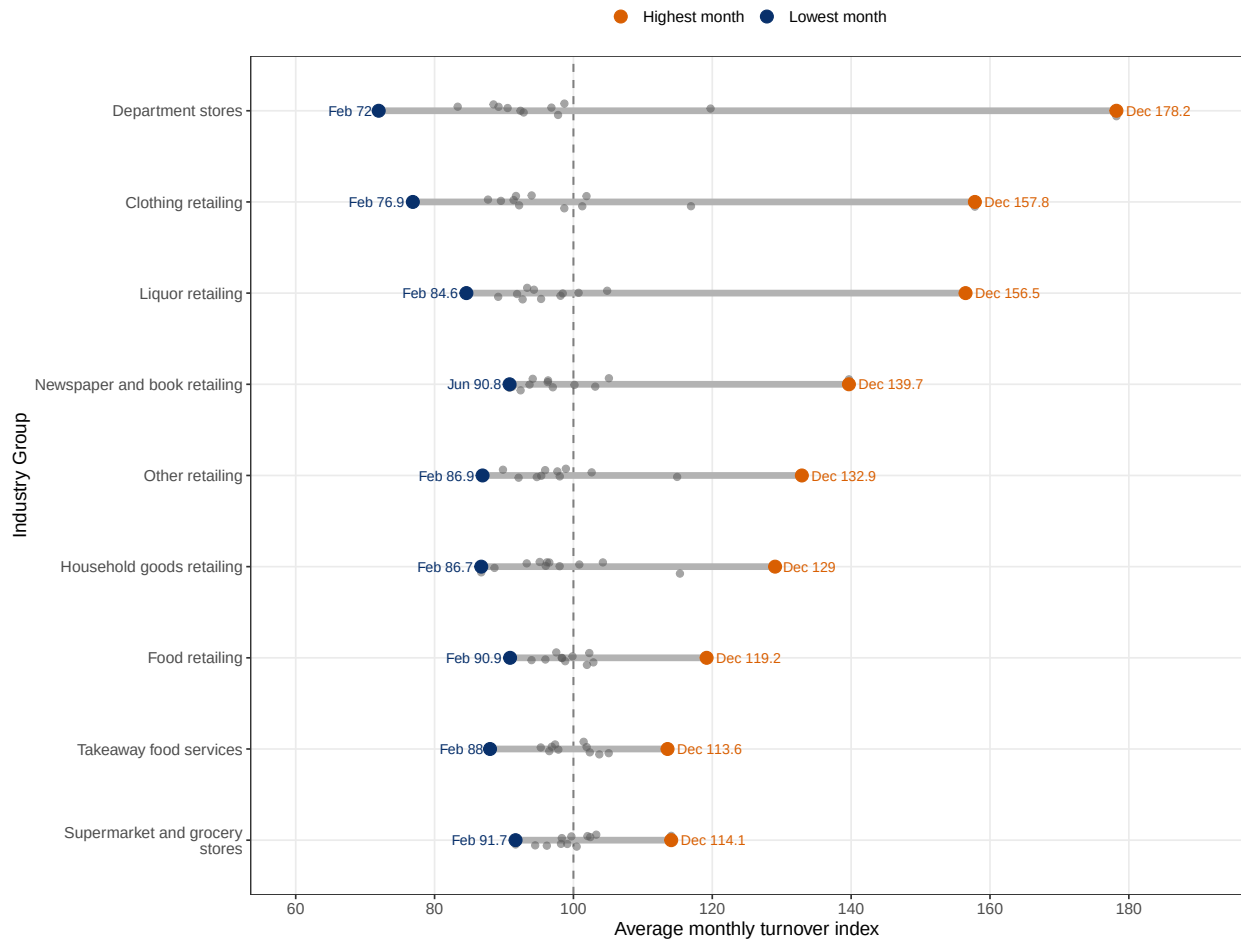
```

) +
geom_text(
  data = seasonal_summary,
  aes(
    x = highest_index,
    y = industry_group,
    label = paste0(highest_month, " ", round(highest_index, 1))
  ),
  hjust = -0.15,
  colour = "#D95F02",
  size = 3.2
) +
scale_colour_manual(
  values = c("Lowest month" = "#08306B", "Highest month" = "#D95F02"),
  name = NULL
) +
scale_x_continuous(breaks = seq(60, 180, by = 20), limits = c(60, 190))
) +
scale_y_discrete(
  labels = function(x) {stringr::str_wrap(x, width = 28)}
) +
labs(
  title = "4. Seasonal Variation in Retail Turnover by Industry",
  subtitle = paste("For each industry and year, the average monthly turnover index = 100"),
  x = "Average monthly turnover index",
  y = "Industry Group",
) +
theme_bw() +
theme(
  text = element_text(size = 12),
  plot.title = element_text(face = "bold", size = 16, hjust = 0.5),
  plot.subtitle = element_text(hjust = 0.5),
  legend.position = "top",
  panel.grid.minor = element_blank()
)

```

4. Seasonal Variation in Retail Turnover by Industry

For each industry and year, the average monthly turnover index = 100



- Which Industries Show the Strongest Seasonal Variation?

Department stores exhibited the widest seasonal gap of all nine industries. Turnover peaked in December at an index of 178.2, roughly 78 points above the annual average, before falling to a low of just 72.0 in February, a difference of 106.2 percentage points. Clothing and liquor retailing followed a similar pattern on a smaller scale, with December peaks of 157.8 and 156.5 respectively, and February lower to 76.9 and 84.6, giving ranges of 80.9 and 71.9 percentage points.

By contrast, supermarket and grocery stores showed the smallest fluctuation, with turnover ranging from 91.7 in February to only 114.1 in December, a gap of just 22.4 percentage points. Takeaway food services and food retailing were similarly steady, with ranges of 25.6 and 28.3 percentage points respectively.

The remaining industries, household goods, other retailing, and newspaper and book retailing, fell between these two extremes, with ranges of roughly 42 to 49 percentage points. Newspaper and book retailing was the only industry not to record its lowest turnover in February, instead reaching its low point in June at an index of 90.8.